

In all the experiments in Figures 3 and 4, using the Gaussian rather than the Laplace kernel results in higher power for our four aggregated tests, the difference is small but notable for the 1-dimensional case while it is large for the 2-dimensional case. For `MMDagg*`, there is no difference in Figure 3 and a small one in Figure 4, as can be seen in the bottom plots. In one dimension, the `median` test performs significantly better when using the Laplace kernel and outperforms the `split` test, which is not the case in all the other settings.

In the bottom plots of Figures 3 and 4, we observe that by aggregating over both Gaussian and Laplace kernels, `MMDagg* Laplace Gaussian` obtains the highest power achieved by either `MMDagg* Laplace` or `MMDagg* Gaussian`. When adding 10 other kernels to the collection (each with 10 bandwidths), `MMDagg* All` retains the same high power, we do not observe a cost in power for considering more kernels. This is only possible due the way we perform the level correction in Section 3.5.

We now discuss the relation between the four weighting strategies for `MMDagg`. Recall from Section 3.5 that a single MMD test with larger associated weight is viewed as more important than one with smaller associated weight in the aggregated procedure. Recall from Section 5.1 that `MMDagg uniform` puts equal weights on every bandwidths, that `MMDagg centred` puts the highest weight on the bandwidth in the middle of the collection, and that `MMDagg increasing` puts the highest weight on the biggest bandwidth while `MMDagg decreasing` puts it on the smallest bandwidth. This allows us to interpret our results.

First, let's consider the case of the collection of bandwidths $\Lambda(-6, -2)$ for both one and two dimensions. We observe that `MMDagg increasing` has the highest power and `MMDagg decreasing` the lowest of the four aggregated tests, this means that putting the highest weight on the biggest bandwidth performs the best while putting it on the smallest bandwidth performs the worst. We can deduce that the most important bandwidth in our collection is the biggest one, which suggests that we should consider a collection consisting of larger bandwidths, say $\Lambda(-4, 0)$. In this case, `MMDagg centred` now obtains the highest power of our four weighting strategies. We can infer that the optimal bandwidth is close to the bandwidths in the middle of our collection. When considering a collection of even larger bandwidths $\Lambda(-2, 2)$, we see the opposite trends to ones observed using $\Lambda(-6, -2)$; `MMDagg decreasing` and `MMDagg increasing` are performing the best and worst of our four tests, respectively. This suggests that a collection consisting of smaller bandwidths than $\Lambda(-2, 2)$ might be more appropriate.

So, comparing our aggregated tests with different weighting strategies gives us some insights on whether the collection we have considered is appropriate, or consists of bandwidths which are either too small or too large. The uniform weighting strategy does not perform the best but it is more robust to changes in the collection of bandwidths than the other strategies. Of course, in practice, if we have access to a limited amount of data, one cannot run a hypothesis test with some parameters, observe the results and then modify those parameters to run the test again. Nonetheless, the interpretation of the results of our different weighting strategies remains an appealing feature of our tests. In practice, we recommend using the parameter-free test `MMDagg* Laplace Gaussian` with its collection of bandwidths chosen adaptively.

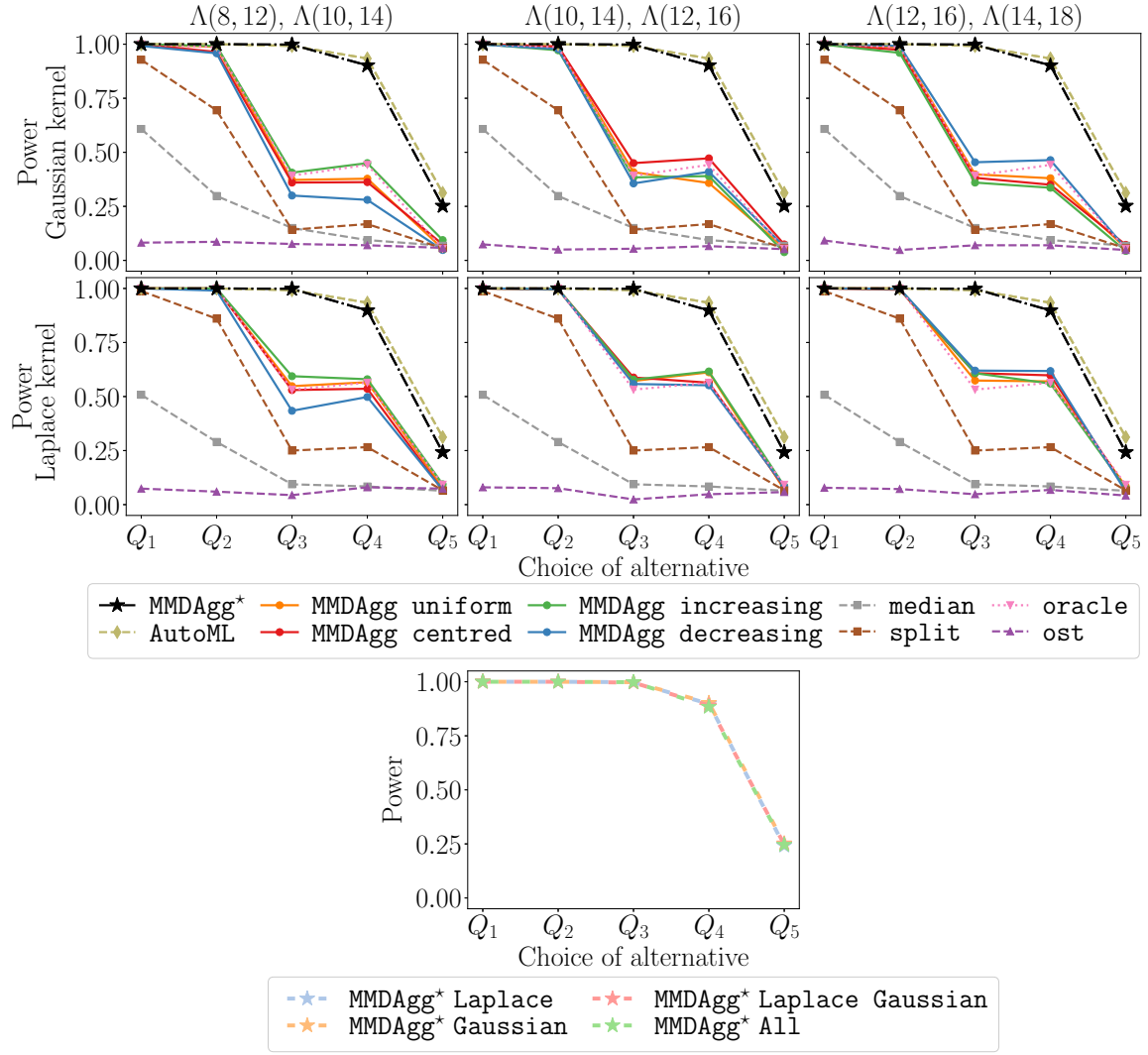


Figure 5: Power experiments with the MNIST dataset using sample sizes $m = n = 500$ with a wild bootstrap.

5.6 Power experiments on the MNIST dataset

Motivated by the experiment considered by Kübler et al. (2020), we consider the MNIST dataset (LeCun et al., 2010) down-sampled to 7×7 images. In Figure 5, we consider 500 samples drawn with replacement from the set $\mathcal{P}$ consisting all 70 000 images of digits

$$\mathcal{P}: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.$$

We test these against 500 samples drawn with replacement against one of the sets

$$\begin{aligned} Q_1 &: 1, 3, 5, 7, 9, \\ Q_2 &: 0, 1, 3, 5, 7, 9, \\ Q_3 &: 0, 1, 2, 3, 5, 7, 9, \\ Q_4 &: 0, 1, 2, 3, 4, 5, 7, 9, \\ Q_5 &: 0, 1, 2, 3, 4, 5, 6, 7, 9, \end{aligned}$$

of respective sizes 35 582, 42 485, 49 475, 56 299 and 63 175. While the samples are in dimension 49, distinguishing images of different digits reduces to a lower-dimensional problem. We consider the Gaussian kernel with the collections of bandwidths $\Lambda(8, 12)$, $\Lambda(10, 14)$ and $\Lambda(12, 16)$ and the Laplace kernel with $\Lambda(10, 14)$, $\Lambda(12, 16)$ and $\Lambda(14, 18)$. As i increases, distinguishing $\mathcal{P}$ from Q_i becomes a more challenging task, which results in a decrease in power.

In the experiments presented in Figure 5 on image data, **MMDAgg*** and **AutoML** achieve the same power and outperform by far all the other tests. This could be due to the fact that the collections for the other aggregated tests consist of very large bandwidths (from $2^8 \lambda_{med}$ to $2^{18} \lambda_{med}$). Nonetheless, we observe that **split** and **oracle**, which consider smaller bandwidths, also perform poorly compared to **MMDAgg*** and **AutoML**.

The four aggregated tests with different weighting strategies often match, or even slightly beat, the performance of **oracle** which uses extra data to select an optimal bandwidth. Moreover, they outperform significantly the two adaptive tests **split** and **ost**, as well as the **median** test.

For **MMDAgg***, using either the Laplace or Gaussian kernel leads to the same performance in this experiment. Furthermore, **MMDAgg*** retains its high power when considering many more kernels as well, as can be seen in the bottom figure of Figure 5 with **MMDAgg* All**. Contrary to the previous experiments, we observe that using the Laplace kernel rather than the Gaussian one results in substantially higher power for the four aggregated tests with different weighting strategies. We recall that in the experiments of Figures 3 and 4, we observed that using a Gaussian kernel leads to higher power. This illustrates that the optimal choice of kernel varies depending on the type of data, as such in practice we recommend using **MMDAgg* Laplace Gaussian** since it is observed to achieve the highest power obtained by either **MMDAgg* Laplace** or **MMDAgg* Gaussian** in those three experiments. The test **MMDAgg* All** also obtains the same power, but as it considers 12 types of kernels, each with 10 bandwidths, it is computationally more expensive.

The pattern we observed in Figures 3 and 4 of having **MMDAgg increasing** and **MMDAgg decreasing** obtaining the highest and lowest power of our four aggregated tests for the collections of smaller and larger bandwidths, respectively, still holds to some extent in Figure 5 but the differences are less significant. For the collection of bandwidths $\Lambda(12, 16)$ with the Laplace kernel, **MMDAgg centred** does not perform the best, it obtains slightly less power than **MMDAgg uniform** and **MMDAgg increasing** which have almost equal power. Following our interpretation, this simply means that, while $\Lambda(12, 16)$ is an appropriate choice of collection, the optimal bandwidth might be slightly larger than $2^{14} \lambda_{med}$.

Note that, except **MMDAgg*** and **AutoML**, each test obtains similar power when trying to distinguish $\mathcal{P}$ from either Q_3 or Q_4 . Recall that Q_3 consists of images of all the digits